# Reasoning test

For admission to Class 9

## About the test

- 15 questions in 15 minutes: 5 each of maths, English and science, mixed together.
- Taken online on a phone or computer, through a link sent to the parent's WhatsApp. The link works for 2 hours.
- Most questions have four options; a few ask for a number to be typed in.
- One mark for a correct answer. A quarter mark off for a wrong multiple-choice answer; nothing off for a wrong type-in answer or a blank.
- Answers can be changed before submitting. If the candidate leaves the page, the timer pauses.

## What the test covers

The questions test reasoning, not memory, using ideas taught up to Class 8. Chapters are from the NCERT (CBSE) books a student joining Class 9 in 2027 will have used; other boards teach the same ideas under their own names.

### Maths

**Class 6 · Ganita Prakash**

- *Patterns in Mathematics* – number patterns and series
- *Perimeter and Area*
- *Prime Time* – primes, factors, HCF and LCM
- *Fractions*

**Class 7 · Ganita Prakash**

- *Working with Fractions* – fractions
- *Finding Common Ground* – primes, factors, HCF and LCM
- *Finding the Unknown* – simple equations

**Class 7 · Curiosity (science)**

- *Measurement of Time and Motion* – speed, distance and time

**Class 8 · Ganita Prakash**

- *A Square and A Cube* – squares and cubes
- *Proportional Reasoning-1* – ratio and proportion
- *Fractions in Disguise* – percentages, profit and loss, discounts, interest
- *Proportional Reasoning-2* – ratio and proportion
- *Tales by Dots and Lines* – averages (mean)

**Reasoning, not a textbook chapter:** Clocks and calendars · Number codes and puzzles · Everyday units · Multi-step word problems · Time and work, pipes and tanks · Problems on ages · Mixtures

### English

**Grammar, vocabulary and reasoning; nothing from the English reader:** Vocabulary: meaning in context, opposites, idioms · Grammar: agreement, prepositions, tenses, linking words · Completing and ordering sentences · Analogies, odd one out, letter series and codes · Reading: inference, tone, assumptions, judging an argument · Logical deduction: statements, arrangements, family relations

### Science

**Class 6 · Curiosity**

- *Exploring Magnets* – magnets
- *A Journey through States of Water* – evaporation and condensation

**Class 7 · Curiosity**

- *Exploring Substances: Acidic, Basic and Neutral* – acids, bases and indicators
- *Electricity: Circuits and their Components* – electric circuits
- *Changes around Us: Physical and Chemical* – physical and chemical changes, rusting
- *Heat Transfer in Nature* – heat and conductors
- *Measurement of Time and Motion* – speed
- *Life Processes in Animals* – respiration
- *Life Processes in Plants* – photosynthesis and plant respiration
- *Light: Shadows and Reflections* – light, shadows and plane mirrors

**Class 8 · Curiosity**

- *Exploring Forces* – floating, sinking and density
- *Pressure, Winds, Storms, and Cyclones* – force and pressure, pressure in liquids
- *Particulate Nature of Matter* – evaporation and condensation
- *How Nature Works in Harmony* – food chains

**Reasoning, not a textbook chapter:** Reading an experiment and fair tests · Applying a principle to a new situation

**Also asked, though the newer NCERT books teach it in a later class:** sound, and the speeds of light and sound

## Sample paper

These questions were written for this sheet. The real test has different questions, in the same style and at a similar level.

1. MATHS

   How many seconds are there in 2.5 hours?

   **A** 150 **B** 900
   **C** 9,000 **D** 15,000

2. ENGLISH

   Choose the word most nearly opposite in meaning to "reluctant".

   **A** eager **B** slow
   **C** afraid **D** unwilling

3. SCIENCE

   Sound cannot travel through

   **A** sea water **B** a steel rail
   **C** still air **D** a vacuum

4. MATHS

   The ratio of boys to girls in a class is 4 : 5. If there are 20 boys, how many girls are there?

   **A** 16 **B** 20
   **C** 24 **D** 25

5. ENGLISH

   Choose the correct word: She has been living in Pune ___ 2019.

   **A** from **B** since
   **C** for **D** by

6. SCIENCE

   A card with a word printed on it is held in front of a plane mirror. In the mirror, the word appears

   **A** upside down **B** with left and right swapped
   **C** larger than on the card **D** smaller than on the card

7. MATHS

   When a number is doubled and then increased by 7, the result is 31. What is the number?

   Type-in answer: _____

8. ENGLISH

   When the coach said the match was "in the bag", he meant that

   **A** the team had packed its kit **B** the result was still uncertain
   **C** victory was almost certain **D** the match had been called off

9. SCIENCE

   Compared with the air we breathe in, the air we breathe out contains

   **A** more carbon dioxide **B** more oxygen
   **C** less water vapour **D** no carbon dioxide at all

10. MATHS

A mother and her daughter are 50 years old together. Five years ago, the mother was 7 times as old as the daughter. How old is the daughter now?

**A** 8
**B** 10
**C** 12
**D** 15

11. ENGLISH

Arrange the sentences in a logical order. P. Asha, their leader, checked the map. Q. The trekkers realised they were lost. R. They reached camp before dark. S. She found the right trail.

**A** PQSR
**B** QSPR
**C** QPRS
**D** QPSR

12. SCIENCE

A sharp knife cuts more easily than a blunt one pushed with the same force, because the sharp edge

**A** makes much more friction with what is cut
**B** pushes with a larger force overall
**C** acts on a smaller area, raising pressure
**D** spreads the force over a larger area

13. MATHS

A scooter worth ₹80,000 loses 10% of its value every year. What will it be worth after 2 years?

**A** ₹64,800
**B** ₹64,000
**C** ₹66,000
**D** ₹72,000

14. ENGLISH

Ali, Bina, Dev, Esha and Tara sit in a row of five. Tara is in the middle, Ali at the left end and Bina next to Ali. Dev is not next to Tara. Who is at the right end?

**A** Dev
**B** Esha
**C** Bina
**D** Cannot be determined

15. SCIENCE

A box weighing 600 N has one face of area 2 $m^2$ and another of 0.5 $m^2$. How many pascals more pressure does it exert resting on the small face than on the large face?

Type-in answer: _____

## Answers

| | | |
|---|---|---|
| 1 | **C. 9,000** | One hour has 60 × 60 = 3,600 seconds, so 2.5 hours has 2.5 × 3,600 = 9,000 seconds. 150 is the number of minutes. |
| 2 | **A. eager** | Reluctant means unwilling or hesitant, so its opposite is eager. "Unwilling" is a word with the same meaning, not the opposite. |
| 3 | **D. a vacuum** | Sound is a vibration passed from particle to particle, so it needs a material. A vacuum has no particles, so sound cannot cross it. |
| 4 | **D. 25** | 20 boys is 4 parts, so one part is 5 students. Girls are 5 parts, which is 25. 16 comes from using the ratio the wrong way round. |
| 5 | **B. since** | "Since" is used with a point in time (since 2019); "for" is used with a length of time (for five years). |
| 6 | **B. with left and right swapped** | A plane mirror forms an image the same size and the same way up, but with left and right swapped, which is why the word reads backwards. |
| 7 | **12** | Work backwards: 31 − 7 = 24, and 24 ÷ 2 = 12. Check: 12 × 2 + 7 = 31. This is a type-in question. |
| 8 | **C. victory was almost certain** | "In the bag" is an idiom meaning that success is as good as secured. It has nothing to do with an actual bag. |
| 9 | **A. more carbon dioxide** | The body uses oxygen to release energy from food and gives off carbon dioxide, so breathed-out air holds more carbon dioxide and less oxygen. |
| 10 | **B. 10** | Five years ago their ages added to 40, in the ratio 7 : 1, so they were 35 and 5. The daughter is now 10 and the mother 40. |
| 11 | **D. QPSR** | The problem comes first (Q), then the response (P), then the solution (S), then the result (R). "She" in S needs Asha named before it. |
| 12 | **C. acts on a smaller area, raising pressure** | Pressure is force divided by area. The same force on the thin edge of a sharp knife gives a much greater pressure, which cuts. |
| 13 | **A. ₹64,800** | After one year it is worth ₹72,000. The second year takes 10% of ₹72,000, which is ₹7,200, leaving ₹64,800. ₹64,000 takes ₹8,000 off twice. |
| 14 | **A. Dev** | Seats 1, 2 and 3 are Ali, Bina and Tara. Dev and Esha take seats 4 and 5; seat 4 is next to Tara, so Dev must be in seat 5. |
| 15 | **900** | On the small face: 600 ÷ 0.5 = 1,200 Pa. On the large face: 600 ÷ 2 = 300 Pa. The difference is 900 Pa. This is a type-in question. |